

DIVYAM NAVIN

Full-Stack Developer | Applied AI / ML

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SUMMARY

Information Technology undergraduate (CGPA 9.68/10) and full-stack developer with hands-on experience building and deploying production-grade web applications (React.js, Node.js, Express, MongoDB) and applied AI/ML systems spanning LLM-based automation, LLM observability and telemetry, multimodal prediction, and computer vision. Comfortable working across the entire stack, from frontend and backend engineering to data pipelines, model experimentation and peer-reviewed ML research, with a track record of shipping real systems across internships, competitions, research, and production projects.

EDUCATION

B.Tech in Information Technology, Fr. C. Rodrigues Institute of Technology 2023 - 2027
Vashi, Navi Mumbai | CGPA: 9.68 / 10 (up to Year 3)

HSC (Science), V.G. Vaze College of Science, Commerce and Arts 2021 - 2023
MHT-CET: 97.5 percentile

RELEVANT COURSEWORK

Data Structures & Analysis, Operating Systems, Computer Networks, Computer Organization & Architecture, Database Management Systems, Software Engineering, Artificial Intelligence, Automata Theory, Data Warehousing & Mining, Cloud Computing, Cryptography & Network Security, Microcontrollers & Embedded Systems, Engineering Mathematics I-IV; Labs: Linux, Python, Data Science, DevOps, Full-Stack Development

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, C, SQL, HTML5, CSS3
Frontend	React.js (React 19), Next.js, Vite, Recharts, Three.js, Responsive Design, UI/UX
Backend	Node.js, Express.js, FastAPI, REST APIs, WebSockets
Databases	PostgreSQL (Supabase), MongoDB, MySQL, NoSQL, ChromaDB (vector store)
AI / ML	PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy, CNNs / Deep Learning, Multimodal Modeling, LLM Integration (Gemini API), Prompt Engineering, RAG, LoRA / QLoRA Fine-Tuning, Local LLM Serving (Ollama, llama.cpp), Sparse Autoencoders / Interpretability, Data Collection and Cleaning
Observability	OpenTelemetry (OTLP), Distributed Tracing & Metrics, SDK Auto-Instrumentation, Anomaly & Incident Detection
Tools & Cloud	Linux, Git/GitHub, Docker, AWS (EC2 GPU), Google Colab, Cloudflare Workers, Vercel, Railway, Render
Concepts	Full-Stack Development, API Integration, Data Preprocessing, Walk-Forward Validation, Cryptography Fundamentals, SEO

EXPERIENCE

AI / Data Intern, Education Domain Jan - Feb 2026
ZetaQ (zetaq.in)

- Built an automated content-processing pipeline that programmatically collected NCERT subject PDFs across multiple standards and routed them through an LLM (Gemini) using structured prompt engineering to extract text and identify key formulas, topics, and concepts.
- Designed prompts to classify and structure the extracted content, generating clean, organized study-material PDFs and cutting down on manual content preparation for the platform.

Full-Stack Web Development Intern 2025
Thinking Engines

- Handled both the frontend and backend of a company website (React.js, Node.js), taking it from design through to deployment.
- Built responsive interfaces and the supporting server-side logic for the full site.

IT Engineer Intern Jun - Jul 2025
Arms Robotics

- Built and deployed the company website (Wave Habitat), integrating real-time hardware-software data flow.
- Engineered a streamlined system architecture for hardware-software interaction that substantially improved data-processing efficiency.
- Supported deployment and testing of production-ready solutions.

- Co-organized flagship entrepreneurship events (Spark-A-Thon, E-Summit 2025 and 2026), led communication, documentation, and team coordination, and mentored early-stage startup ideas.

PROJECTS

LLM Observatory: Observability Tool for LLM Workloads

2026 - Present

Major Project, in collaboration with the emerging Open Tech Foundation (eOTF) | Team of 4

- Building an OpenTelemetry-native observability platform for LLM applications: FastAPI collector with an OTLP receiver, PostgreSQL (Supabase), a React dashboard and a drop-in Python SDK for 15 LLM providers.
- Traces latency, tokens, cost and errors per call, with prompt versioning, quality scoring, prompt-injection and PII detection, hallucination (grounding) checks, and incident detection with root-cause ranking.
- Validated on ~6,500 real traced calls: 7.7 microsecond SDK overhead per call, 0.93 AUROC on grounding checks, and a LoRA fine-tuned 1.7B model that beat Meta's Prompt Guard 2 at injection detection (F1 0.94).

Multimodal Stock Market Prediction System

2025 - Aug 2026

Personal Project (completed)

- Built a near real-time market analysis system that combines technical indicators, news and sentiment signals, and fundamentals to generate directional predictions and buy/sell guidance.
- Reached around 60 to 62% directional accuracy on short-term (roughly one month) horizons, built in Python with an ensemble prediction layer over engineered multimodal features.

Image-to-Biomass Estimation

2025

CSIRO Problem Statement

- Developed a multimodal model that combines tabular features with a CNN over imagery to predict vegetation biomass, and also predicted NDVI directly from the imagery.
- Built the full pipeline covering preprocessing, modeling, and evaluation, and applied regularization to reduce overfitting on a limited dataset.

EduSage, AI-Assisted Education Management Platform

2024 - 2025

Academic Project

- Architected a full-stack education platform (React.js, Node.js, MongoDB) to streamline academic workflows.
- Integrated an LLM-powered module (Gemini API and prompt engineering) into the interview and quiz process for a more intuitive, responsive experience.

Wave Habitat, Real-Time Hardware Monitoring System

2025

Arms Robotics

- Built a real-time monitoring dashboard (Node.js and API integration) that bridges embedded hardware with a web interface for live data interaction.
- Implemented scalable backend logic to handle high-frequency data streams from embedded systems, reducing manual troubleshooting time.

Portfolio Website

2025

portfolio-dn-mocha.vercel.app

- Designed and built an interactive 3D portfolio (JavaScript, Three.js, custom GLSL shaders) where a single particle system morphs into a unique scene for each internship and project, with adaptive quality for mobile devices.
- (Previous version: designed and deployed a high-performance personal portfolio (React.js, Node.js, Express, Cloudflare Workers) with a responsive UI and high availability, divyam-n-portfolio.vercel.app)

PUBLICATIONS

BiasLens: Sparse Autoencoder Feature Probes for Interpretable Social Bias Detection in Small Language Models

2026

Accepted at ICSIT 2026 (IEEE-sponsored) | Mechanistic interpretability, PyTorch

- Proposed a Cohen's-d based Bias Feature Score over sparse-autoencoder features as a lightweight, interpretable bias probe, evaluated on CrowS-Pairs with multi-seed robustness and StereoSet zero-shot transfer.

AWARDS AND ACHIEVEMENTS

- Received multiple Letters of Recommendation for technical, academic, and strategic contributions.
- Ranked among the top performers in the Information Technology branch.